INTRODUCTION TO ICT

Lesson 2

Representation of Data in computer

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I. Numeral systems

- **Numeral system**:
  - Is a set of symbols and rules, using which we can represent and determine the value of numbers.
  - Each numeral system uses a finite set of characters/symbols/digits. The number of characters/symbols/digits is called the base or radix of the numeral system, normally denoted by b
  - E.g.: In the decimal system, we use ten digits: 0,1,2,…,9, hence b =10

I. Numeral systems

- Mathematically, we can represent a number in any numeral system
- In computer science, we usually using following counting systems:
  - Decimal system
  - Binary system
  - Octal system
  - Hexadecimal system

Decimal system

- The decimal system (with b = 10) using 10 digits:

0, 1, 2, 3, 4, 5, 6, 7, 8, 9

- In the decimal system, using n digits we can represent 10^n different values, ranging from 0 to 10^n-1:

00...000 = 0

....

99...999 = 10^n-1

Decimal system

- Supposing that a number A which is represented in the decimal system as:

  $A = a_n\, a_{n-1} \ldots a_1\, a_0 \,.\, a_{-1}\, a_{-2} \ldots a_{-m}$

  $\rightarrow$ The value of A is determined as:

$$A = a_n 10^n + a_{n-1} 10^{n-1} + \ldots + a_1 10^1 + a_0 10^0 + a_{-1} 10^{-1} + \ldots + a_{-m} 10^{-m}$$

$$A = \sum_{i=-m}^{n} a_i 10^i$$

Decimal system

- E.g. 1: The value of the number 5246 is determined as:

  $5246 = 5 \times 10^3 + 2 \times 10^2 + 4 \times 10^1 + 6 \times 10^0$

- E.g. 2: The value of the number 254.68 is determined as:

  $254.68 = 2 \times 10^2 + 5 \times 10^1 + 4 \times 10^0 + 6 \times 10^{-1} + 8 \times 10^{-2}$

b-base system

- In general, a b-base numeral system (b >=2, integer):
  - Using b digits, ranging from 0 to b-1
  - A number $N_{(b)}$ in the b-base numeral system is represented as:

    $N_{(b)}=a_{n-1}a_{n-2}\ldots a_1a_0.a_{-1}a_{-2}\ldots a_{-m}$
  - In the above representation, n digits is used to represent integer part and m digits for fraction

b-base system

– The value of number $N_{(b)}$ is determined as:

$$N_{(b)} = a_n.b^n + a_{n-1}.b^{n-1} + a_{n-2}.b^{n-2} + ... + a_1.b^1 + a_0.b^0 + a_{-1}.b^{-1} + a_{-2}.b^{-2} + ... + a_{-m}.b^{-m}$$

hay là:

$$N_{(b)} = \sum_{i=-m}^{n} a_i.b^i$$

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The binary system

- Base b = 2
- Using 2 digits: 0 and 1
- Binary digit (0 or 1) is called a ***bit*** (**bi**nary digi**t)**

  E.g.: bit 0, bit 1
- Bit is the smallest information unit

The binary system

- Using n bits we can represent 2^n different value, ranging from 0 to 2^n -1:

  00...000 (2)= 0

  ...

  11...111 (2)= 2^n - 1

- E.g.: Using 2 bits we can represent 4 values: 00 (=0); 01 (=1); 10 (=2); 11 (=3)

The binary system

- Supposing that a number A which is represented in the binary system as:

$A = a_n\, a_{n-1} \ldots a_1\, a_0 \,.\, a_{-1}\, a_{-2} \ldots a_{-m}$

,in which a_i is binary digit (0,1), then value of A is determined as:

$$A = a_n 2^n + a_{n-1} 2^{n-1} + \ldots + a_1 2^1 + a_0 2^0 + a_{-1} 2^{-1} + a_{-2} 2^{-2} + \ldots + a_{-m} 2^{-m}$$

$$A = \sum_{i=-m}^{n} a_i 2^i$$

The binary system

- E.g.: The value of the binary number 1101001.1011 is determined as:

$$1101001.1011_{(2)} = 2^6 + 2^5 + 2^3 + 2^0 + 2^{-1} + 2^{-3} + 2^{-4}$$

$$= 64 + 32 + 8 + 1 + 0.5 + 0.125 + 0.0625$$

$$= 105.6875_{(10)}$$

Plus/Minis in the binary system

- Plus:

  1+0=0+1=1;

  0+0=0;

  1+1=10;

- Minus

  1-1=0;

  0-0=0;

  1-0=1;

  0-1=1; (borrow 1)

Example: Plus

```
  1  0  1
+ 1  1  1
---------
1  1  0  0
```

Example: Minus

```
    1 1 0 0
-     1 1 1
-------------
    0 1 0 1
```

The octal system

- Base b = 8
- Using 8 digits: 0,1,2,3,4,5,6,7
- With n digits we can represent 8^n different values, ranging from 0 to 8^n -1 as:

  00...000 = 0

  ...

  77...777 = 8^n -1

The octal system

- Supposing that an A number which is represented in the octal system as:

$$A = a_n\, a_{n-1} \ldots a_1\, a_0 \,.\, a_{-1}\, a_{-2} \ldots a_{-m}$$

with a_i is digit in the octal system, then value of A is determined as:

$$N_{(b)} = a_n.b^n + a_{n-1}.b^{n-1} + a_{n-2}.b^{n-2} + \ldots + a_1.b^1 + a_0.b^0 + a_{-1}.b^{-1} + a_{-2}.b^{-2} + \ldots + a_{-m}.b^{-m}$$

hay là:

$$N_{(b)} = \sum_{\ldots}^{n} a_i.b^i$$

with b = 8.

The octal system

- E.g.: Determine the value of

$235\,.\,64_{(8)}$:

$235\,.\,64_{(8)} = \mathbf{2}x8^{2} + \mathbf{3}x8^{1} + \mathbf{5}x8^{0} + \mathbf{6}x8^{-1} + \mathbf{4}x8^{-2} =$

$157.\,8125(10)$

The hexadecimal system

- Base b = 16
- Using 16 digits and characters:
  0,1,2,3,4,5,6,7,8,9,A,B,C,D,E,F

  in which characters:

  A, B, C, D, E, F

  represent respected number in decimal system

  10, 11, 12, 13, 14, 15.

Hệ thập phân	Hệ nhị phân	Hệ mười sáu
0	0000	0
1	0001	1
2	0010	2
3	0011	3
4	0100	4
5	0101	5
6	0110	6
7	0111	7
8	1000	8
9	1001	9
10	1010	A
11	1011	B
12	1100	C
13	1101	D
14	1110	E
15	1111	F

The hexadecimal system

- Supposing that an A number which is represented in hexadecimal system as:

  $A = a_n\ a_{n-1}\ \ldots\ a_1\ a_0\ .\ a_{-1}\ a_{-2}\ \ldots\ a_{-m}$

  with a_i is digit in hexadecimal system, the value of A is determined as:

$$A = a_n 16^n + a_{n-1}16^{n-1} + \ldots + a_1 16^1 + a_0 16^0 + a_{-1}16^{-1} + a_{-2}16^{-2} + \ldots + a_{-m}16^{-m}$$

$$A = \sum_{i=-m}^{n} a_i 16^i$$

The hexadecimal system

- E.g.: Determine value of $34F5C.12D_{(16)}$:

$34F5C.12D_{(16)} =$

$3x16^4 + 4x16^3 + 15x16^2 + 5x16^1 + 12x16^0$

$+? \qquad = 216294(10) +?$

$?=1/16 + 2/16/16 +13/16/16/16=0.07348..$

Convert a number from decimal system to other numeral system

- In general a number N in the decimal system ($N_{(10)}$) consists of 2 part: Integer and Fraction.
- Converting a number from decimal system to other numeral system including 2 steps:
  - Convert integer part from decimal system to b-base system
  - Convert fraction part from decimal system to b-base system

Convert a number from decimal system to other numeral system

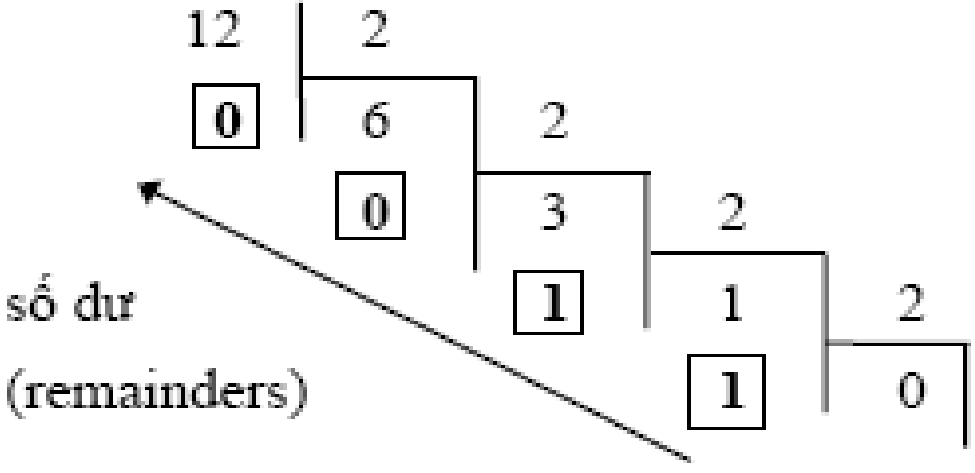
- Convert integer part:
  - **Step 1:** Divide the integer part of $N_{(10)}$ by b, we obtain T_1, remainder is d_1.
  - **Step 2:** if T_1 <> 0, Divide T_1 by b, we obtain T_2, remainder is d_2

    ....
  - **Step n:** $T_n = 0$, remainder is d_n => stop
  - Result number N(b) obtained by remainders in above steps as:

    $N_{(10)} = d_n d_{n-1} \ldots d_{1\,(b)}$

Convert a number from decimal system to other numeral system

– E.g.: Convert integer part of number $12.6875_{(10)}$ to binary system:



Kết quả: $12_{(10)} = 1100_{(2)}$

Convert a number from decimal system to other numeral system

- Convert fraction part of a number in decimal system to other system:
  - Step 1: Multiple fraction part of $N_{(10)}$ with b, we obtain a number in form of $x_1.y_1$ (in which x in integer, y is fraction)
  - Step 2: If $y_1 <> 0$, multipe $0.y_1$ with b, we obtain $x_2.y_2$
  - Step n: If $y_{n-1} <> 0$, multiple $0.y_{n-1}$ with b, we obtain $x_n.0$, => stop
  - Result number is:

    $0,x_1x_2\ldots x_n$

Convert a number from decimal system to other numeral system

– E.g.: Convert fraction part of number $12.6875_{(10)}$ to binary system:

Ví dụ 3.11: $0.6875_{(10)} = ?_{(2)}$

phần nguyên của tích

0. 6875	x 2 =	1	375	phần thập phân của tích
0. 3750	x 2 =	0	. 75	
0. 75	x 2 =	1	. 5	
0. 5	x 2 =	1	. 0	

Kết quả: $0.6875_{(10)} = 0.1011_{(2)}$

Examples

1. Decimal → Binary 124.75 = ?

   1111100.11

2. Decimal → Binary 65.125 = ?

   1000001.001

3. Binary → Hexa: 1011 1110 0110 = ?

BE6

4. Hexa → Binary: 3E8 = ?

0011 1110 1000

5. Hexa → Decimal: 3A8C = ?

= 14988

Hệ thập phân
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15

- 124.75
- 124 /2 = 62 , 0
- 62 /2 = 31 , 0
- 31 / 2 = 15 , 1
- 15 / 2 = 7 , 1
- 7 /2 = 3 , 1
- 3 / 2 = 1 , 1
- 1 / 2 = 0 , 1
- 124 = 1111100

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- 0.75 x2 = 1 , 5
- 0.5 x 2 = 1 , 0
- -.75 = 11
- 124.75 = 1111100.11

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Represent data in computer

- All data using in computer must be encoded into binary numbers
- Data in computer is classified into 2 types:
  - Basic data
  - Structured data

Basic data

- Integer
  - Unsigned integer: Using binary.
  - Signed integer: Using the 2's complement encoding.
- Real:
  - Using floating point number
- Character:
  - Using character encode: ASCII, Unicode, …

Structured data

- Is a set of basic data which is structured in a specific manner.
- E.g.: Array, string, record, …
- Study more in other courses like Programming language.

Information Unit

- **Bit** (Binary digit): là đơn vị thông tin nhỏ nhất, nhận 1 trong 2 giá trị nhị phân là 0 hoặc 1.
- **Byte**: chuỗi 8 bit.
- **Kilobyte** (KB), 1KB = 2^{10} Byte = 1024 Byte
- **Megabyte** (MB), 1MB = 2^{10} KB = 2^{20} Byte = 1,048,576 Byte
- **Gigabyte** (GB), 1GB = 2^{10} MB = 2^{30} Byte = 1,073,741,824 Byte
- **Terabyte** (TB), 1TB = 2^{10} GB = 2^{40} Byte
- **Petabyte** (PB), 1PB = 2^{10} TB = 2^{50} Byte
- **Exabyte** (EB), 1EB = 2^{10} PB = 2^{60} Byte

Represent data in computer

1. Representation of integer number
2. Representation of real number
3. Representation of character

Representation of integer number

- Using a set of bits.
- For signed integer, we use the first bit (Most significant bit) to represent “-” sign. We call this bit is signed bit.

Representation of unsigned integer

- Supposing that a number A is represented using n bits as:

$$a_{n-1}a_{n-2}...a_3a_2a_1a_0$$

then, value of A is determined as:

$$A = a_{n-1}2^{n-1} + a_{n-2}2^{n-2} + ... + a_1 2^1 + a_0 2^0$$

$$A = \sum_{i=0}^{n-1} a_i 2^i$$

- Using n bit to represent unsigned integer, we can represent values from 0 to 2^n-1

Examples

- E.g. 1: Representing unsigned integer A and B numbers using 8 bits:

A = 45 B = 156

A = 45 = 32 + 8 + 4 + 1 = $2^5 + 2^3 + 2^2 + 2^0$

→ A = 0010 1101

B = 156 = 128 + 16 + 8 + 4 = $2^7 + 2^4 + 2^3 + 2^2$

→ B = 1001 1100

Examples

- E.g. 2: Given representation of unsigned integer numbers X and Y, determine values of X and Y:

  X = 0010 1011

  Y = 1001 0110

  $X = 0010\ 1011 = 2^5 + 2^3 + 2^1 + 2^0$

  $= 32 + 8 + 2 + 1 = 43$

  $Y = 1001\ 0110 = 2^7 + 2^4 + 2^2 + 2^1$

  $= 128 + 16 + 4 + 2 = 150$

Representation of signed integer

- Given representation of signed integer A as:

$$a_{n-1}a_{n-2}...a_2a_1a_0$$

then, value of A is determined as:

$$A = -a_{n-1}2^{n-1} + \sum_{i=0}^{n-2} a_i 2^i$$

- Using n bits, values of A is in: $[-2^{n-1}, 2^{n-1}-1]$

10000…000

……..

01111…111

Representation of signed integer

- Given representation of signed integer A as:

$$a_{n-1}a_{n-2}\ldots a_2a_1a_0$$

- If $a_{n-1} := 0$:
  - A is positive number
  - $a_{n-2}\ldots a_2a_1a_0$ represent value of A

$$A = \sum_{i=0}^{n-2} a_i 2^i$$

  - Values of A is in: $[0, 2^{n-1}-1]$

Representation of signed integer

- Given representation of signed integer A as:

$$a_{n-1}a_{n-2}\ldots a_2a_1a_0$$

- If a_{n-1}:= 1:
  - A is negative number
  - Value of A is determined as:

$$A = -2^{n-1} + \sum_{i=0}^{n-2} a_i 2^i$$

  - Values of A is in: $[-2^{n-1}, -1]$

Representation of signed integer

- E.g.: Determine value of signed integers A and B:

A = 0101 0110

B = 1101 0010

$A = 2^6 + 2^4 + 2^2 + 2^1 = 64 + 16 + 4 + 2 = +86$

$B = -2^7 + 2^6 + 2^4 + 2^1 =$

$= -128 + 64 + 16 + 2 = -46$

Representation of signed integer

Convert a signed integer number (+/-) from decimal system to binary system

Representation of signed integer

- Using the 2's complement encoding
- Given an unsigned integer A which is represented in the binary system using n bits, then:
  - The 1's complement encoding of A = $(2^n - 1) - A$
  - The 2's complement encoding of A = $2^n - A$
- E.g.:
  - given A = 0110
  - The 1's complement encoding of A = $(2^4 - 1) - 0110 = 1001$
  - The 2's complement encoding of A = $2^4 - 0110 = 1010$

Representation of signed integer

- A signed integer number –A is represented by the 2's complement encoding of A
- E.g.: Represent signed number A = -70 using 8 bits

A = 70 = 0100 0110

1's comp.: 1011 1001

\+ 1

2's comp.: 1011 1010

Hence: -70 => 1011 1010

Representation of Real number

- General rule:
  - In order to represent real number in computer, we use Floating point number

Representation of Real number

- A floating point number has the form of:

  $X = M * R^E$

  in which:
  - M is mantissa
  - R is radix (base), usually be 2 or 10.
  - E is exponent
- Given R, to represent a real number X, we need to determine M and E (as integer number)

Representation of Real number

- E.g.: Using R = 10, given two real numbers N1 and N2 represented in floating point with:
  - M1 = -15 và E1 = +12
  - M2 = +314 và E2 = -2

  Determine the values of N1 and N2?

Operations on floating point real number

- Given two floating point real numbers:
  - $N1 = M1 \times R^{E1}$ and $N2 = M2 \times R^{E2}$
- Add/Subtract:

$$N1 \pm N2 = (M1 \times R^{E1-E2} \pm M2) \times R^{E2},$$

- Multiple and divide

$$N1 \times N2 = (M1 \times M2) \times R^{E1+E2}$$

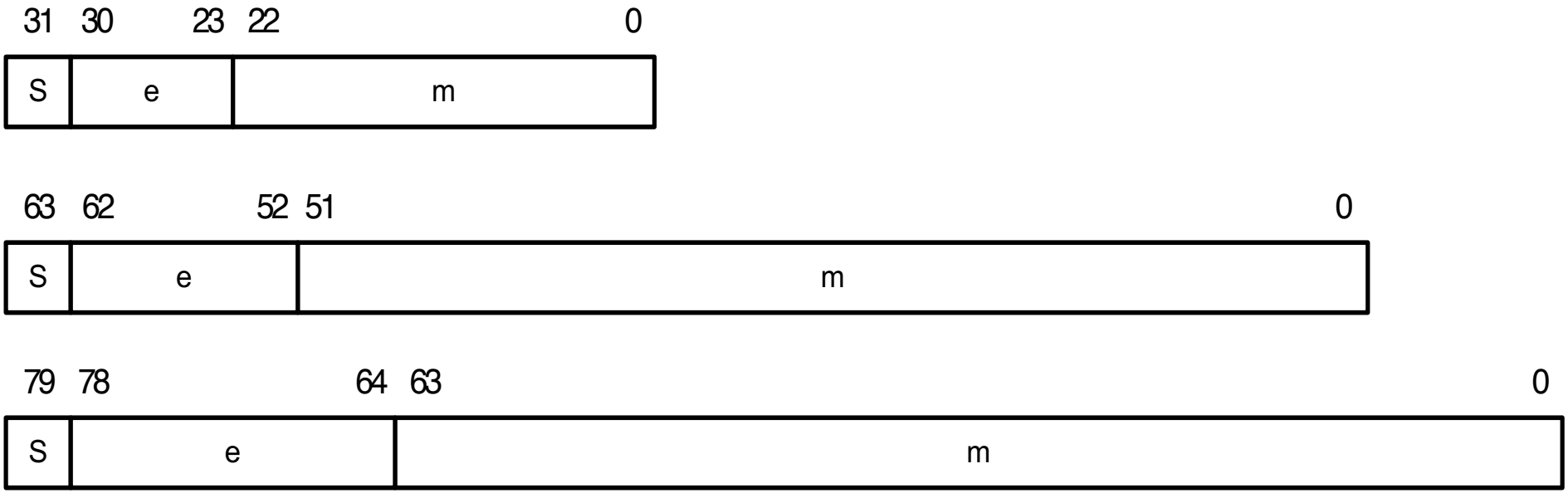
$$N1 / N2 = (M1 / M2) / R^{E1-E2}$$

IEEE 754/85 format

- Is used to represent Floating point number
- Radix R = 2
- Consist of 3 types:
  - Single precision, using 32-bit
  - Double precision, using 64-bit
  - Extended double precision, using 80-bit

IEEE 754/85 format

Encoding format



IEEE 754/85 format

- S is signed bit: S=0 for positive and and S=1 for negative number.
- e is excess of exponent E: E = e – b

  where b is bias:
  - With 32-bit : b = 127, then E = e - 127
  - With 64-bit : b = 1023, then E = e - 1023
  - With 80-bit : b = 16383, then E = e – 16383
- m is fraction part of M: M = 1.m

IEEE 754/85 format



- Value of X is determined as follow:

$$X = (-1)^{S} \times 1.m \times 2^{e-b}$$

Example

- E.g.1: Given X which is represented in IEEE 754 32 bit:

1100 0001 0101 0110 0000 0000 0000 0000

Determine value of X.

- E.g.2: Given Y which is represented in IEEE 754 32 bit:

0011 1111 1000 0000 0000 0000 0000 0000

Determine value of Y.

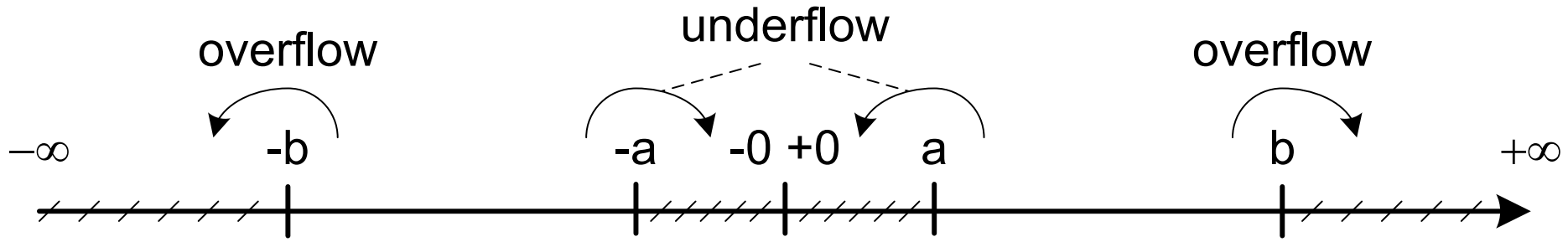
Example

- E.g. 3: Given a real number X = 6.375. Represent X in IEEE 754 using 32 bit

Note

- If all bits of e are 0 and all bits of m are 0, then X = $\pm$ 0
- If all bits of e are 1 and all bits of m are 0, then X = $\pm \infty$
- If all bits of e are 1 and at least one bit of m is 1, the X is not a number (NaN)

Representation range



- 32 bit: $a = 2^{-127} \approx 10^{-38}$ $b = 2^{+127} \approx 10^{+38}$
- 64 bit: $a = 2^{-1023} \approx 10^{-308}$ $b = 2^{+1023} \approx 10^{+308}$
- 80 bit: $a = 2^{-16383} \approx 10^{-4932}$ $b = 2^{+16383} \approx 10^{+4932}$

Representation of characters

- **General rule:**
  - Convert characters to binary.
  - The number of bits used to encode each character depend on encoding set.
  - E.g. :
    - ASCII uses 8 bits
    - Unicode uses 16 bits.

ASCII

- Proposed by ANSI (American National Standard Institute)
- ASCII uses 8 bit, hence it can represent 256 characters
- Character has code in range: $00_{16} \div FF_{16}$

ASCII Chart

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0	NUL	SOH	STX	ETX	EOT	ENQ	ACK	BEL	BS	HT	LF	VT	FF	CR	SO	SI
1	DLE	DC1	DC2	DC3	DC4	NAK	SYN	ETB	CAN	EM	SUB	ESC	FS	GS	RS	US
2	SPC	!	"	#	$	%	&	'	(	)	*	+	,	-	.	/
3	0	1	2	3	4	5	6	7	8	9	:	;	<	=	>	?
4	@	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
5	P	Q	R	S	T	U	V	W	X	Y	Z	[	\	]	^	_
6	`	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o
7	p	q	r	s	t	u	v	w	x	y	z	{	|	}	~	DEL

Physical Device Controls: Format Effectors

ASCII TABLE

Decimal	Hex	Char	Decimal	Hex	Char	Decimal	Hex	Char	Decimal	Hex	Char
0	0	[NULL]	32	20	[SPACE]	64	40	@	96	60	`
1	1	[START OF HEADING]	33	21	!	65	41	A	97	61	a
2	2	[START OF TEXT]	34	22	"	66	42	B	98	62	b
3	3	[END OF TEXT]	35	23	#	67	43	C	99	63	c
4	4	[END OF TRANSMISSION]	36	24	$	68	44	D	100	64	d
5	5	[ENQUIRY]	37	25	%	69	45	E	101	65	e
6	6	[ACKNOWLEDGE]	38	26	&	70	46	F	102	66	f
7	7	[BELL]	39	27	'	71	47	G	103	67	g
8	8	[BACKSPACE]	40	28	(	72	48	H	104	68	h
9	9	[HORIZONTAL TAB]	41	29	)	73	49	I	105	69	i
10	A	[LINE FEED]	42	2A	*	74	4A	J	106	6A	j
11	B	[VERTICAL TAB]	43	2B	+	75	4B	K	107	6B	k
12	C	[FORM FEED]	44	2C	,	76	4C	L	108	6C	l
13	D	[CARRIAGE RETURN]	45	2D	-	77	4D	M	109	6D	m
14	E	[SHIFT OUT]	46	2E	.	78	4E	N	110	6E	n
15	F	[SHIFT IN]	47	2F	/	79	4F	O	111	6F	o
16	10	[DATA LINK ESCAPE]	48	30	0	80	50	P	112	70	p
17	11	[DEVICE CONTROL 1]	49	31	1	81	51	Q	113	71	q
18	12	[DEVICE CONTROL 2]	50	32	2	82	52	R	114	72	r
19	13	[DEVICE CONTROL 3]	51	33	3	83	53	S	115	73	s
20	14	[DEVICE CONTROL 4]	52	34	4	84	54	T	116	74	t
21	15	[NEGATIVE ACKNOWLEDGE]	53	35	5	85	55	U	117	75	u
22	16	[SYNCHRONOUS IDLE]	54	36	6	86	56	V	118	76	v
23	17	[END OF TRANS. BLOCK]	55	37	7	87	57	W	119	77	w
24	18	[CANCEL]	56	38	8	88	58	X	120	78	x
25	19	[END OF MEDIUM]	57	39	9	89	59	Y	121	79	y
26	1A	[SUBSTITUTE]	58	3A	:	90	5A	Z	122	7A	z
27	1B	[ESCAPE]	59	3B	;	91	5B	[	123	7B	{
28	1C	[FILE SEPARATOR]	60	3C	<	92	5C	\	124	7C	|
29	1D	[GROUP SEPARATOR]	61	3D	=	93	5D	]	125	7D	}
30	1E	[RECORD SEPARATOR]	62	3E	>	94	5E	^	126	7E	~
31	1F	[UNIT SEPARATOR]	63	3F	?	95	5F	_	127	7F	[DEL]

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ASCII

- 95 displayable characters has code:20_{16} ÷ $7E_{16,}$ in which:
  - 'A' ÷ 'Z' has code: 41_{16} ÷ $5A_{16}$
  - 'a' ÷ 'z' has code: 61_{16} ÷ $7A_{16}$
  - '0' ÷ '9' has code: 30_{16} ÷ 39_{16}